

READING PASSAGE 3

El Nino and Climate Variability

A In recent years, el nino and climate variability has attracted considerable attention from both the academic community and the general public. This growing interest reflects a broader recognition of the importance of this topic for understanding some of the most pressing challenges facing contemporary society.

B Historically, scholarly understanding of this topic was limited by both theoretical frameworks and available methodologies. Early researchers operated within relatively narrow conceptual boundaries, and their findings, while groundbreaking for their time, often failed to capture the full complexity of the phenomena they sought to explain. The development of new research tools and analytical techniques in the latter half of the twentieth century fundamentally transformed the landscape, enabling scholars to investigate questions that had previously been considered beyond the reach of empirical inquiry.

C Contemporary research has identified several critical factors that determine outcomes in this domain. The interplay between environmental conditions, technological capabilities, and social organisation creates a complex adaptive system in which small changes can produce disproportionately large effects. Understanding these dynamics requires analytical approaches that move beyond simple cause-and-effect reasoning to embrace the principles of complexity science and systems thinking.

D A substantial body of empirical evidence supports these theoretical frameworks. A landmark study conducted by researchers at leading international institutions analysed data spanning a thirty-year period and involving participants from more than forty countries. The findings revealed a consistent pattern of results that held across diverse cultural and economic contexts, suggesting that the identified mechanisms operate with considerable universality. The magnitude of the observed effects was particularly striking, with the key variables accounting for a substantial proportion of the total variance in outcomes.

E Despite the substantial progress that has been made, significant challenges and controversies remain. Critics have raised important methodological concerns regarding the generalisability of findings derived from studies conducted in specific contexts. The predominance of research carried out in Western, educated, industrialised, rich, and democratic societies raises questions about the applicability of current theories to populations with different cultural, economic, and political circumstances. Furthermore, the reliance on cross-sectional designs in much of the existing literature limits the capacity to draw strong causal inferences.

F The implications of this body of research extend well beyond the academy. As societies grapple with increasingly complex challenges, the insights generated by scholarly inquiry in this domain have the potential to inform more effective and equitable policy responses. Realising this potential, however, will require sustained investment in both fundamental research and the translational infrastructure needed to convert knowledge into action.

G In conclusion, the study of el nino and climate variability represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Questions 1–2

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 1 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.
- 2 The author notes that researchers have called for more _____ approaches to investigation.

Question 3

Choose the correct letter, A, B, C or D.

- 3 The meta-analyses cited in the passage reveal that:
 - A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations

Question 4

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 4 The landmark study discussed in the passage involved participants from multiple countries.

Questions 5–6

Choose the correct letter, A, B, C or D.

- 5 What finding emerged from the landmark study discussed in the passage?
 - A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations
- 6 What finding emerged from the landmark study discussed in the passage?
 - A. inconsistent results across cultures
 - B. a consistent pattern across diverse contexts
 - C. no measurable effects
 - D. effects limited to Western countries

Question 7

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

7 Policy-makers have fully implemented all research recommendations.

Question 8

Choose the correct letter, A, B, C or D.

- 8 According to the text, what role do structural factors play?
- A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations

Questions 9–10

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 9 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.
- 10 The author believes that all research questions in this field have been answered.

Questions 11–12

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 11 Dynamic _____ processes can amplify or dampen initial effects in complex systems.
- 12 The landmark study analysed data spanning a period of _____ years.

Question 13

Choose the correct letter, A, B, C or D.

- 13 The author suggests that early studies in this field were limited by:
- A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations